

Wk 2 Notes

1. How are drones used for aerial imagery purposes?
 - a. Can be used to survey land, and UAVs stitches photos together taken onboard.
 - b. Agriculture
 - c. Oblique vs Nadir (directly above) imagery
2. How is OpenCV used for computer vision purposes? How do we use it as a club?
 - a. OpenCV: Library used for object detection
3. What are some pre-made alternatives to writing our own aerial imagery code, and how do they work? What are the drawbacks?
 - a. OpenCV: Remove color channel, shrink/stretch, blur, sharpen, and modify images.
 - b. Using APIs that have pre-made tools for you
 - c. OpenDroneMaps: simple, nice APIs, can avoid coding (?),
4. What are the next steps forward as a club?
 - a. SLAM (Simultaneous Localization and Mapping)
 - b. Optical Flow
 - c. Structure from Motion (getting a terrain map from image data)
 - d. Applying drone imagery

Other Notes

OpenCV: Library used for object detection

- Feature Mapping to Stitch Images or to make a map (3D or 2D).
 - OpenDroneMap (ODM) or OpenCV
 - ODM: has APIs, can give cool models and data. Not good for real-time on drone computers.

Hint for Stitches Option:

I'll be Needin' Stitches: Hints

- You're going to need `cv2.VideoCapture(video)` to handle the video data. From there, you can `cap.read()`.
 - You can downsample (sample fewer images) with modular arithmetic!
- Use `cv2.SIFT_create()` or `cv2.ORB_create()` to detect keypoints and descriptors.
- Then you can match them with a feature matcher (also in OpenCV)!
- Try making the images match with `cv2.warpAffine` and then overlaying.
- OR you can try using the builtin `cv2.Stitcher`—your choice!!

